AP Vidyut Recruitment Exam 23rd to 27th Aug 2026

Hall Ticket Number	APVR0031994
Participant Name	SIREESHA
Test Center Name	iON Digital Zone iDZ 2 Chinamushidiwada
Test Time	9:00 AM - 12:00 PM
Test Date	26/08/2026
Subject	Electrical Engineering

Section : **General Intelligence and Reasoning**

Q.1 **24 is related to 9 following a certain logic. Following the same logic, 120 is related to 33. To which of the following is 84 related to, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)**

Ans
✔ 1. 24
✘ 2. 36
✘ 3. 32
✘ 4. 134

Question ID : **441009585207**
Chosen Option : **2**

Q.2 **Refer to the given letter series and answer the question that follows.**

(Left) X Z O W B Y M F H G U K R E S I A Q T C D (Right)

How many letters are there in the English alphabetical series between the letter which is sixth from the right end and seventh from the left end of the series?

Ans
✘ 1. Two
✔ 2. Three
✘ 3. Five
✘ 4. Four

Question ID : **441009797062**
Chosen Option : **1**

Q.3 **Based on the English alphabetical order, three of the following four letter-clusters are alike in a certain way and thus form a group. Which letter-cluster DOES NOT belong to that group? (Note: The odd one out is not based on the number of consonants/vowels or their position in the letter-cluster.)**

Ans
✗ 1. LHJ
✗ 2. HDF
✗ 3. OKM
✓ 4. FCD

Question ID : **441009781802**
Chosen Option : **4**

Q.4 **Seven people, A, B, C, D, E, F and G are sitting in a row, facing north. Only five people sit between F and G. E sits to the immediate left of G. Only one person sits between E and A. C sits at some place to the left of D but at some place to the right of B. Who sits second to the right of C?**

Ans
✓ 1. D
✗ 2. A
✗ 3. E
✗ 4. B

Question ID : **441009814183**
Chosen Option : **1**

Q.5 **Based on the English alphabetical order, three of the following four are alike in a certain way and thus form a group. Which is the one that does not belong to that group?**
(Note: The odd one out is not based on the number of consonants/vowels or their position in the letter cluster)

Ans
✓ 1. KPM
✗ 2. NSQ
✗ 3. RWU
✗ 4. CHF

Question ID : **441009765196**
Chosen Option : **1**

Section : **General awareness**

Q.6 **Who was the President of the Indian National Congress during its Eleventh Session in 1895?**

Ans
✗ 1. Dadabhai Naoroji
✗ 2. Gopal Krishna Gokhale
✗ 3. Bal Gangadhar Tilak
✓ 4. Surendranath Banerjee

Question ID : **4410092115976**
Chosen Option : **1**

Q.7 **The Automated Weather Station (AWS) installed at a glacial lake in Sikkim in 2023 is primarily intended to provide early warning against which hazard?**

Ans
1. Prolonged drought caused by a weak monsoon
2. Widespread frost associated with a western disturbance
3. Coastal storm surge during the retreating monsoon
4. Glacial Lake Outburst Flood triggered by intense rainfall

Question ID : **4410092115981**
Chosen Option : **1**

Q.8 **Which of the following mountain ranges is primarily composed of ancient metamorphic rocks, considered one of the oldest geological formations in India?**

Ans
1. The Western Ghats
2. The Vindhya Range
3. The Aravalli Range
4. The Himalayas

Question ID : **4410092258049**
Chosen Option : **1**

Q.9 **According to the Consolidated FDI Policy circular of 2020, what is the threshold for classifying an investment in a listed Indian company as Foreign Direct Investment (FDI) rather than Foreign Portfolio Investment (FPI)?**

Ans
1. 1% or more of the post-issue paid-up equity capital
2. 5% or more of the post-issue paid-up equity capital
3. 25% or more of the post-issue paid-up equity capital
4. 10% or more of the post-issue paid-up equity capital

Question ID : **4410092200869**
Chosen Option : **2**

Q.10 **Who was appointed as the Chairperson of the National Statistical Commission (NSC) in June 2026?**

Ans
1. T. C. A. Anant
2. Saibal Chattopadhyay
3. Bimal Kumar Roy
4. Pronab Sen

Question ID : **4410091453928**
Chosen Option : **3**

Q.11 **Which major administrative change did the British implement in India following the Revolt of 1857?**

Ans
1. India was divided into new independent kingdoms
2. The East India Company’s rule continued
3. Indian rulers were completely removed
4. The British Crown took direct control of India

Question ID : **441009244778**
Chosen Option : **2**

Q.12 **Mallika Sarabhai is best known for her expertise in which two classical dance forms?**

Ans
1. Bharatanatyam and Kuchipudi
2. Kathakali and Mohiniyattam
3. Odissi and Kathak
4. Bharatanatyam and Kathakali

Question ID : **441009284032**
Chosen Option : **1**

Section : **English**

Comprehension:

Read the passage carefully and answer the questions that follow.

Semiconductor manufacturing has emerged as a strategic priority as geopolitical tensions and supply-chain disruptions have exposed the vulnerabilities of excessive dependence on a handful of chip-producing economies. Consequently, governments are no longer treating semiconductor fabrication merely as an industrial activity but as an instrument of economic resilience and technological sovereignty. India's recent Semicon 2.0 initiative extends this logic by coupling financial incentives with investments in design capabilities, fabrication facilities, advanced packaging, and critical materials rather than pursuing fabrication in isolation. Such an integrated approach acknowledges that competitiveness in the semiconductor ecosystem depends less on constructing individual fabrication plants than on cultivating an interconnected network of research, skilled talent, suppliers, and downstream industries. Accordingly, long-term strategic success will be determined not by manufacturing capacity alone but by the sustained integration of innovation, industrial collaboration, and resilient supply-chain governance.

SubQuestion No : 13

Q.13 **Which of the following best summarizes the passage?**

Ans
1. Semiconductor policies primarily seek to increase the number of fabrication plants because manufacturing capacity alone determines technological leadership and long-term industrial competitiveness.
2. India's semiconductor strategy concentrates mainly on financial subsidies for fabrication projects because research, packaging, and supplier ecosystems contribute comparatively little to competitiveness.
3. Modern semiconductor initiatives emphasise building an integrated ecosystem combining manufacturing, innovation, skills, and supply chains to strengthen economic resilience and technological capability.
4. Geopolitical tensions have permanently disrupted semiconductor trade, compelling countries to replace international cooperation entirely with complete domestic production and market insulation.

Question ID : **4410092413408**
Chosen Option : **3**

Comprehension:

Read the passage carefully and answer the questions that follow.

Semiconductor manufacturing has emerged as a strategic priority as geopolitical tensions and supply-chain disruptions have exposed the vulnerabilities of excessive dependence on a handful of chip-producing economies. Consequently, governments are no longer treating semiconductor fabrication merely as an industrial activity but as an instrument of economic resilience and technological sovereignty. India's recent Semicon 2.0 initiative extends this logic by coupling financial incentives with investments in design capabilities, fabrication facilities, advanced packaging, and critical materials rather than pursuing fabrication in isolation. Such an integrated approach acknowledges that competitiveness in the semiconductor ecosystem depends less on constructing individual fabrication plants than on cultivating an interconnected network of research, skilled talent, suppliers, and downstream industries. Accordingly, long-term strategic success will be determined not by manufacturing capacity alone but by the sustained integration of innovation, industrial collaboration, and resilient supply-chain governance.

SubQuestion No : 14

Q.14 Which option most accurately describes the organisational structure of the passage?

Ans
✗ 1. It introduces a policy initiative, compares it with international semiconductor programmes, evaluates their relative effectiveness, and recommends the superior framework.

✗ 2. It outlines the historical evolution of semiconductor manufacturing, examines successive technological advances, and predicts future industrial transformation.

✗ 3. It presents an economic problem, critiques existing fiscal incentives, evaluates statistical evidence, and argues against government intervention in semiconductor production.

✓ 4. It identifies a contemporary challenge, explains its broader implications, illustrates an emerging policy response, and concludes with a general principle about industrial competitiveness.

Question ID : **4410092413409**
Chosen Option : **1**

Q.15 Complete the following sentence by choosing the correct option.
During the debate, the speaker ____ addressed each criticism, demonstrating both logic and composure.

Ans
✗ 1. silently

✓ 2. systematically

✗ 3. emotionally

✗ 4. confusedly

Question ID : **4410091711179**
Chosen Option : **1**

Q.16 Change the given sentence into passive voice.
Nancy has written this letter.

Ans
✓ 1. This letter has been written by Nancy.

✗ 2. This letter was written by Nancy.

✗ 3. This letter had been written by Nancy.

✗ 4. This letter will have been written by Nancy.

Question ID : **441009484088**
Chosen Option : **1**

Q.17 Substitute one word for the given underlined segment by choosing from the following options.
The court's decision was intended to formally declare a previous judicial ruling or law as null and void due to its inconsistency with the new constitution.

Ans
1. litigate
2. codify
3. abrogate
4. adjudicate

Question ID : **4410091713147**
Chosen Option : **1**

Section : **Quantitative aptitude**

Q.18 There are 48 members in group A, 60 members in group B and 42 members in group C. All the members of these groups went to a restaurant. The average amount spent on each member of group A, B and C is ₹350, ₹383 and ₹460, respectively. The total average amount (in ₹) spent per member is:5555

Ans
1. 390
2. 399
3. 398
4. 394

Question ID : **441009582459**
Chosen Option : **1**

Q.19 20% of a number is more than 20% of 700 by 160. The number is:3859

Ans
1. 1550
2. 1600
3. 1350
4. 1500

Question ID : **441009855141**
Chosen Option : **4**

Q.20 The volume of a cuboidal block of silver is 1,536 cm^3. If its dimensions are in the ratio of 4:3:2, find the cost of gold polishing its entire surface at the rate of ₹2.50 per cm^2.

Ans
1. ₹2,140
2. ₹2,260
3. ₹2,080
4. ₹2,380

Question ID : **4410091556698**
Chosen Option : **4**

Q.21 Three friends P, Q and R started a business by investing a sum of money in the ratio of 8 : 4 : 7. After 6 months Q withdraws half of his capital. If the sum invested by P is ₹31000, out of a total annual profit of ₹24000 find the share of Q in profit.

Ans
1. ₹4835
2. ₹4000
3. ₹3225
4. ₹2230

Question ID : **441009548027**
Chosen Option : **4**

Q.22 In a 2700 m circular race, Raman finishes one round in 30 seconds and Mohan finishes one round in 75 seconds. How many different meeting points are there on circumference if they are running in a same direction?

Ans
1. 5
2. 1
3. 8
4. 3

Question ID : **441009538575**
Chosen Option : **2**

Q.23 Two taps can fill a cistern in 2 hours and 44 hours, respectively. A third tap can empty it in 44 hours. How long (in hours) will it take to fill half of the empty cistern, if all of them are opened together?

Ans
1. 2
2. 3
3. 1
4. 4

Question ID : **441009518586**
Chosen Option : **3**

Q.24 Ten squares of different sizes are pasted on a chart paper adjacent to each. If the first ten multiples of three represent the sides in cm of the ten squares, find the area (in cm^2) of the chart paper covered.

Ans
1. 3465
2. 1155
3. 385
4. 10395

Question ID : **44100913483**
Chosen Option : **1**

Q.25 **Find the simple interest (in ₹) if a sum of ₹500 is borrowed for 7.5 years at 12% per annum rate of interest.**

Ans
1. 430
2. 550
3. 500
4. 450 ✔

Question ID : **441009852060**
Chosen Option : **4**

Section : **Computer knowledge**

Q.26 **While editing a presentation in MS PowerPoint, a user needs to adjust the dimensions of an inserted image without removing it from the slide. Which feature is most appropriate for this task?**

Ans
1. Animation Pane
2. Resize handles ✔
3. Format Painter
4. Crop tool

Question ID : **4410092272417**
Chosen Option : **2**

Q.27 **A company switches from a traditional hard disk drive system to a hybrid storage architecture that combines volatile and non-volatile memory modules for its database operations. Which outcome most accurately reflects the effect of this change on the system's data handling and performance?**

Ans
1. All data is permanently erased when power is lost, regardless of memory type used.
2. Data retrieval becomes slower because the system checks multiple memory types for every operation.
3. System stability is unaffected, as memory configuration has no impact on data security or access speed.
4. Frequently accessed information is retrieved more quickly due to volatile memory, while long-term data remains secure in non-volatile modules. ✔

Question ID : **4410092276829**
Chosen Option : **4**

Q.28 **A research lab wants to interconnect its climate sensors, control systems, and local data servers within a single building to allow rapid data exchange and device management. Which network configuration, if chosen, would present the most significant risk of signal collision and performance bottlenecks as the number of connected devices increases, assuming all devices share equal bandwidth allocation and operate simultaneously?**

Ans
1. Implementing a single shared communication channel for all device connections within the building ✔
2. Creating device subgroups with independent bandwidth allocation per group
3. Assigning bandwidth dynamically to devices based on real-time usage needs
4. Setting up multiple isolated channels for different building segments

Question ID : **4410092276899**
Chosen Option : **3**

Q.29 A logistics company uses an Excel function to automatically assign delivery routes based on a dynamic list of driver locations that updates every hour. Which function would most reliably ensure the route assignments always correspond to the most recently updated driver data when the spreadsheet recalculates, even if drivers are added or removed from the list during the day?

Ans

✗ 1. OFFSET with static range references fixed at setup

✓ 2. INDEX with dynamic named ranges referencing the updated list

✗ 3. VLOOKUP with a fixed table array defined at initial configuration

✗ 4. INDIRECT with hard-coded cell references for driver data

Question ID : **4410092276341**
Chosen Option : **2**

Q.30 Which option in MS Word 2021 allows you to change the alignment of text within a table cell?

Ans

✓ 1. Table Layout tab

✗ 2. Review tab

✗ 3. Home tab

✗ 4. Table Design tab

Question ID : **4410092307500**
Chosen Option : **4**

Section : **Electrical Engineering**

Q.31 In a medium-length transmission line, which characteristic of the line is mainly responsible for the real power loss that occurs due to current flow and results in reduced transmission efficiency?

Ans

✗ 1. Line inductance

✗ 2. Power factor variation

✓ 3. Conductive property of the line material

✗ 4. Line capacitance

Question ID : **4410091549706**
Chosen Option : **4**

Q.32 In atomic hydrogen electric arc welding, the electric arc struck between two tungsten electrodes in an atmosphere of hydrogen. What is the function of hydrogen in the process?

Ans

✗ 1. Hydrogen atmosphere acts as electrode itself for the welds

✓ 2. Hydrogen acts as a cooling agent for the glowing tungsten electrode tips

✗ 3. Hydrogen causes the weld to form oxides and nitrides for high quality of weld

✗ 4. Hydrogen acts as a non-conducting atmosphere for controlling the arc.

Question ID : **4410091557171**
Chosen Option : **3**

Q.33 As the load on a synchronous motor increases, the inverted V-curve corresponding to that load shifts ________:

Ans

✗ 1. Upward because the motor draws more reactive power.

✗ 2. To the left, indicating that lesser field current is now required for feeding the load at unity power factor.

✓ 3. To the right, requiring higher field current to feed the load at unity power factor.

✗ 4. Downward due to increased copper losses.

Question ID : **4410091549531**
Chosen Option : **3**

Q.34 A single-phase half-wave controlled rectifier with freewheeling diode is supplying an RL load with R = 10 Ω and L = 10mH. If the input supply is 220 V, 50 Hz and firing angle is 60°, then what is the average output voltage of the rectifier?

Ans

✓ 1. $\frac{165\sqrt{2}}{\pi}$ V

✗ 2. $\frac{110\sqrt{2}}{\pi}$ V

✗ 3. $\frac{330\sqrt{2}}{\pi}$ V

✗ 4. $\frac{330}{\pi}$ V

Question ID : **4410091565576**
Chosen Option : **1**

Q.35 A basic voltage doubler circuit consists of ________.

Ans

✗ 1. Transistors only

✗ 2. Resistors and inductors

✓ 3. Diodes and capacitors

✗ 4. MOSFETs and resistors

Question ID : **4410091553474**
Chosen Option : **3**

Q.36 What is the primary distinguishing feature that a Gate Turn-Off (GTO) thyristor differs from a conventional Silicon Controlled Rectifier (SCR)?

Ans

✗ 1. It turns off by reversing the anode-to-cathode voltage (natural commutation).

✓ 2. It can be turned off by applying a large, high-amplitude negative current pulse to the gate terminal.

✗ 3. It can be turned off by applying a large, sustained positive current to the gate terminal.

✗ 4. It can be turned off by applying a momentary positive pulse to the gate terminal.

Question ID : **4410091550325**
Chosen Option : **2**

Q.37 In a cathode ray oscilloscope, the voltage of an unknown signal is obtained by observing:

Ans
1. Time base setting
2. Vertical deflection per division
3. Input impedance of CRO
4. Horizontal sweep frequency

Question ID : **4410091547204**
Chosen Option : **2**

Q.38 While designing the foundation for a substation structure, which of the following loads must be considered to ensure safety and stability?

Ans
1. Only wind loads
2. Only vertical loads
3. Only lateral loads
4. Both vertical and lateral loads

Question ID : **4410091813280**
Chosen Option : **4**

Q.39 The mechanical force on a current-carrying conductor in a magnetic field depends on which of the following?

Ans
1. Permittivity of the medium
2. Magnetic flux density
3. Frequency of electrical supply of the conductor
4. Resistance of the conductor

Question ID : **4410091545945**
Chosen Option : **2**

Q.40 In a single-phase half-wave circuit (SPHWC), the discontinuous current occurs when:

Ans
1. Inductance is large
2. Firing angle = 0
3. Supply voltage is high
4. Current becomes zero before next cycle

Question ID : **4410091564081**
Chosen Option : **4**

Q.41 Adding an Integral controller typically makes the system ___________.

Ans
1. More sluggish
2. Faster
3. Overshoot-free
4. Noise-free

Question ID : **4410091554040**
Chosen Option : **2**

Q.42 In a digital system, which gate would you use to reverse the logical state of a signal before it enters another logic circuit?

Ans
✗ 1. Transmission Gate
✗ 2. AND Gate
✓ 3. NOT Gate
✗ 4. OR Gate

Question ID : **4410091792640**
Chosen Option : **3**

Q.43 Which of the following statements are valid reasons for limiting the heat-production rate at very high frequencies in dielectric heating?

i. At very high frequencies, tuning the inductance to resonate with the circuit capacitance becomes difficult.
ii. At very high frequencies, achieving a uniform voltage distribution across the material becomes very difficult.
iii. The penetration depth of electromagnetic waves decreases significantly at very high frequencies.
iv. The dielectric constant of the material continues to increase without limit as the frequency increases.

Ans
✗ 1. ii, iii, iv only
✗ 2. i, ii, iv only
✗ 3. i, iii, iv only
✓ 4. i, ii, iii only

Question ID : **4410091555356**
Chosen Option : **2**

Q.44 Which of the following parameter/s most strongly influences the quality of a flash butt weld?

Ans
✗ 1. Type of shielding gas used
✗ 2. Electrode coating thickness
✓ 3. Welding current and flashing time
✗ 4. Frequency of the supply voltage

Question ID : **4410091557401**
Chosen Option : **3**

Q.45 During prolonged rainfall, the soil supporting a pedestal transformer foundation becomes fully saturated, leading to an increase in pore water pressure. Which soil parameter controlling the stability of the foundation is most likely to decrease?

Ans
✗ 1. Unit weight of soil
✓ 2. Shear strength of soil
✗ 3. Permeability of soil
✗ 4. Compressibility of soil

Question ID : **4410091813189**
Chosen Option : **1**

Q.46 **In a single-phase half-wave controlled converter with an RLE load, the thyristor turns off at an angle ωt = β after switching on the input supply at t = 0. If the minimum firing angle is $\theta_1 = \sin^{-1}(E/V_m)$, and the supply angular frequency is ω rad/s, which of the following correctly gives the circuit turn-off time?**

Ans

✔ 1. $t_c = \dfrac{2\pi + \theta_1 - \beta}{\omega}$

✗ 2. $t_c = \dfrac{\beta - \theta_1}{\omega}$

✗ 3. $t_c = \dfrac{\pi + \theta_1 - \beta}{\omega}$

✗ 4. $t_c = \dfrac{\pi + \beta - \theta_1}{\omega}$

Question ID : **4410091563174**
Chosen Option : **1**

Q.47 **A 3-phase, 5 MVA industry substation is feeding the load at 0.8 power factor lagging. However, due to electricity rules, this industry has to maintain its power factor at unity and for that the industry decided to install a capacitor bank. What will be the per phase rating of the capacitor bank?**

Ans

✗ 1. 3 MVAr

✗ 2. 4 MVAr

✗ 3. 1.33 MVAr

✔ 4. 1 MVAr

Question ID : **4410091565363**
Chosen Option : **4**

Q.48 **When the quantity of conducting material in an auto-transformer is reduced, the required window area is also reduced, which ultimately leads to ________________.**

Ans

✗ 1. A larger core length and higher core weight

✗ 2. An increase in the core area

✔ 3. A smaller core length and lower core weight

✗ 4. No change in the core dimensions

Question ID : **4410091545698**
Chosen Option : **3**

Q.49 **Line conductance represents:**

Ans

✔ 1. Leakage current at the insulators

✗ 2. Charging current

✗ 3. Corona effect

✗ 4. Magnetic flux

Question ID : **4410091561457**
Chosen Option : **1**

Q.50 **A V-to-I converter is also called a/an _______.**

Ans

1. Comparator

2. Integrator

3. Differentiator

4. Transconductance amplifier

Question ID : **4410091553516**
Chosen Option : **3**

Q.51 **A circuit containing both RC and RL components exhibits:**

Ans

1. Purely resistive response

2. Only capacitive behavior

3. Only inductive behavior

4. First-order or second-order response depending on topology

Question ID : **4410091557938**
Chosen Option : **4**

Q.52 **Consider a connected graph with 'n' nodes and 'b' branches. The incidence matrix 'A' is defined such that each row corresponds to a node and each column to a branch. Which of the following statements is always true for the incidence matrix of a connected graph?**

Ans

1. The rank of the incidence matrix is always equal to the number of branches b

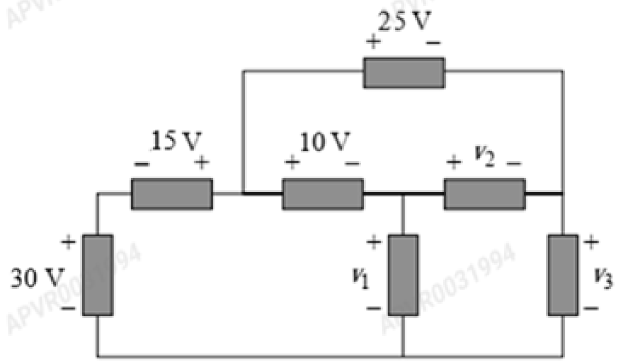
2. The incidence matrix is symmetric for all undirected graphs

3. The product $A \cdot A^T$ yields a diagonal matrix with branch degrees on the diagonal

4. The sum of all rows of the incidence matrix is a zero vector

Question ID : **4410091565535**
Chosen Option : **1**

Q.53 For the following circuit, find the values of v_1 , v_2 , and v_3 , respectively.



Ans
- ✗ 1. v_1 = 35 V, v_2 = 15 V, v_3 = -20 V
- ✗ 2. v_1 = -35 V, v_2 = 15 V, v_3 = 20 V
- ✗ 3. v_1 = 35 V, v_2 = -15 V, v_3 = 20 V
- ✓ 4. v_1 = 35 V, v_2 = 15 V, v_3 = 20 V

Question ID : **4410091554790**
Chosen Option : **2**

Q.54 Source transformation is mainly used to:

Ans
- ✓ 1. Simplify complex circuit networks
- ✗ 2. Convert nonlinear elements into linear ones
- ✗ 3. Reduce total current
- ✗ 4. Increase circuit voltage

Question ID : **4410091557582**
Chosen Option : **1**

Q.55 If a substation foundation is constructed in a chemically aggressive soil environment, which of the following characteristics of concrete should be prioritized to ensure durability?

Ans
- ✗ 1. High cement content to increase paste volume in the mix
- ✗ 2. Fast setting properties to accelerate early strength development
- ✗ 3. Low slump to decrease workability during placement operations
- ✓ 4. Low permeability to limit ingress of moisture and harmful chemicals

Question ID : **4410091816713**
Chosen Option : **2**

Q.56 **If the block diagram of a control system consists of forward gain G1 then G2 in series and the feedback path has H(s) (negative) sampling output and subtracting at input, the equivalent closed-loop transfer function of the control system is ________.**

Ans ✔ 1. (G1G2)/(1+G1G2H)

✗ 2. (G1+G2)/(1+G1G2H)

✗ 3. (G1+G2)/(1+H)

✗ 4. (G1.G2)/(1+H)

Question ID : **4410091565164**
Chosen Option : **1**

Q.57 **Which stability check ensures that a pedestal transformer foundation does not rotate due to lateral forces?**

Ans ✔ 1. Overturning Stability

✗ 2. Sliding stability

✗ 3. Settlement control

✗ 4. Bearing capacity check

Question ID : **4410091813199**
Chosen Option : **1**

Q.58 **An ideal inductor has zero current passing through it before t = 0. At t = 0, it is suddenly connected to a DC current source. How will the inductor behave right after this moment ($t = 0^+$)?**

Ans ✗ 1. It will behave like an ideal current source.

✗ 2. It will behave like a non-ideal current source (with some internal resistance).

✗ 3. It will behave like a short circuit.

✔ 4. It will behave like an open circuit.

Question ID : **4410091565492**
Chosen Option : **4**

Q.59 **A digital voltmeter operates by converting the analog voltage into:**

Ans ✔ 1. Digital form using an ADC

✗ 2. Electrostatic force

✗ 3. Frequency modulation

✗ 4. Magnetic deflection

Question ID : **4410091547293**
Chosen Option : **1**

Q.60 **For a fully transposed transmission line, zero-sequence impedance is usually:**

Ans
1. Zero
2. Equal to positive sequence impedance
3. Equal to negative sequence impedance
4. Higher than both positive and negative sequence impedance

Question ID : **4410091561259**
Chosen Option : **1**

Q.61 **Which of the following methods is used to control the magnitude of the fundamental component of the output voltage in a Single-Pulse Modulated inverter?**

Ans
1. Changing the width of the single pulse, δ.
2. Varying the frequency of the carrier signal.
3. Implementing a filter network at the output.
4. Adjusting the DC input voltage, V_s.

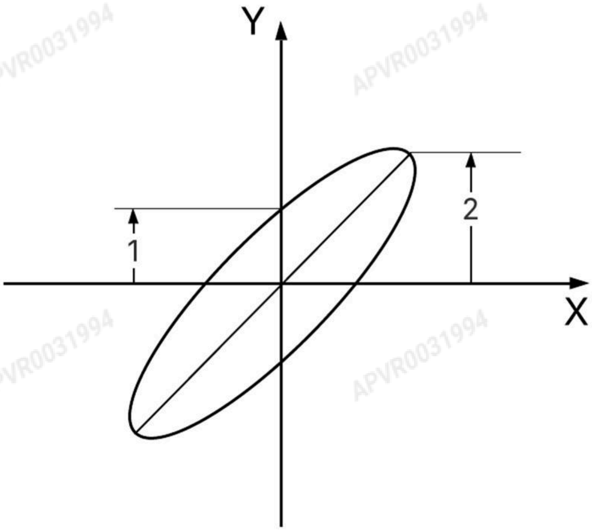
Question ID : **4410091550358**
Chosen Option : **1**

Q.62 **Two-reaction theory is applied to:**

Ans
1. DC generators
2. Salient pole synchronous machines
3. Induction motors
4. Cylindrical rotor alternators

Question ID : **4410091555063**
Chosen Option : **4**

Q.63 **A CRO displays a Lissajous pattern, as shown below, for voltages of the same frequency but out of phase, which are connected to the Y and X plates of the oscilloscope. If the spot generating the pattern moves in an anticlockwise direction, what is the phase difference between the two signals?**



Ans

✗ 1. +60°

✗ 2. +30°

✓ 3. −30°

✗ 4. −60°

Question ID : **4410091537551**
Chosen Option : **4**

Q.64 **What is the primary function of the Maxwell's Inductance-Capacitance type bridge?**

Ans

✗ 1. To measure the unknown capacitance in terms of known inductance.

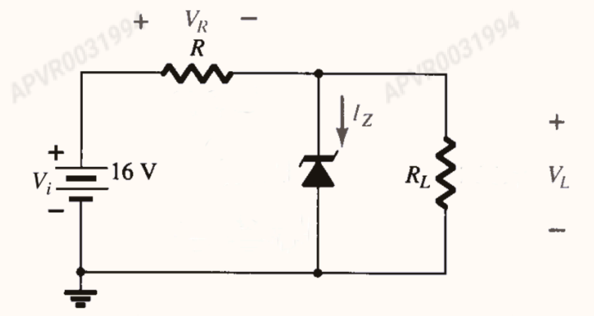
✓ 2. To measure the unknown self-inductance (L_x) in terms of a standard capacitor (C_1).

✗ 3. To measure very high-quality factor (Q) coils (Q > 10).

✗ 4. To measure the unknown frequency of the AC source.

Question ID : **4410091544744**
Chosen Option : **2**

Q.65 **For the Zener diode network shown below, the input voltage is V_i = 16 V, series resistance R = 1 kΩ, load resistance R_L = 1.2 kΩ and Zener breakdown voltage V_Z = 10 V. Assume ideal Zener diode behavior. The voltage V_R across the series resistor R is______.**



Ans
1. 16 V
2. 10 V
3. 6 V
4. 7.27 V

Question ID : **4410091563107**
Chosen Option : **3**

Q.66 **A synchronous MOD-12 counter is designed using T flip-flops. The minimum number of T flip-flops required is ______.**

Ans
1. 6
2. 12
3. 5
4. 4

Question ID : **4410091576293**
Chosen Option : **1**

Q.67 **The 'loading effect', in which a measuring instrument changes the quantity being measured, is explicitly referred to as a type of which error subcategory?**

Ans
1. Gross Error
2. Environmental Error
3. Instrumental Error
4. Observational Error

Question ID : **4410091544767**
Chosen Option : **2**

Q.68 **An engineer is tasked with transmitting a high-fidelity audio signal over long distances. Considering distortion and noise immunity, which modulation type would be most appropriate and why?**

Ans
1. AM modulation, as it uses less power for long-distance transmission.
2. AM modulation, due to its lower complexity.
3. FM modulation, as it requires less bandwidth.
4. FM modulation, due to its high noise immunity and fidelity.

Question ID : **4410091805688**
Chosen Option : **1**

Q.69 **In AC arc welding machines, the welding transformer is designed to have:**

Ans
1. Extremely high frequency output
2. Equal primary and secondary voltages
3. High secondary voltage and low current
4. Low secondary voltage and high current

Question ID : **4410091562239**
Chosen Option : **4**

Q.70 **At the resonant frequency, the response curve of a parallel circuit shows ____________ and the sharpness of the resonance curve in a parallel resonance circuit is determined by its ____________.**

Ans
1. Maximum current, Power rating
2. Minimum current, Q-factor
3. Sudden voltage drops, Q-factor
4. Zero impedance, Frequency tolerance

Question ID : **4410091546047**
Chosen Option : **2**

Q.71 **How does a monostable multivibrator react if retriggered during its high output state?**

Ans
1. Force a second pulse of reduced amplitude
2. Ignore the trigger and continue existing pulse
3. Drive the op-amp into slew-rate limiting mode
4. Retrigger immediately and extend the pulse

Question ID : **4410091557705**
Chosen Option : **4**

Q.72 **Autotransformers are preferred in:**

Ans
1. Protection circuits
2. Power Amplifiers
3. Large transformation ratio
4. Starting of induction motors

Question ID : **4410091557248**
Chosen Option : **4**

Q.73 **During concreting in waterlogged soil, which factor primarily increases the risk of washout and loss of cement paste?**

Ans
- ✗ 1. Low cement content
- ✗ 2. Floating of coarse aggregates
- ✗ 3. Insufficient mixing time
- ✓ 4. Excessive water movement

Question ID : **4410091792949**
Chosen Option : **1**

Q.74 **If an open-loop system has one pole in the right-half of the s-plane, the system is ________.**

Ans
- ✗ 1. Stable
- ✓ 2. Unstable
- ✗ 3. Conditionally stable
- ✗ 4. Marginally stable

Question ID : **4410091554152**
Chosen Option : **2**

Q.75 **For a 20-bus power system with 30 transmission lines, what is the sparsity of the Y-bus matrix?**

Ans
- ✓ 1. 0.80
- ✗ 2. 0.85
- ✗ 3. 0.50
- ✗ 4. 0.65

Question ID : **4410091572716**
Chosen Option : **4**

Q.76 **Which of the following options is correct for the system with transfer function T(s) = (s-4)/s(s+3)?**

Ans
- ✗ 1. It has 1 pole and two zeros and the system is minimum phase system.
- ✗ 2. It has 2 poles and one zero and the system is minimum phase system.
- ✗ 3. It has 1 pole and two zeros and the system is non-minimum phase system.
- ✓ 4. It has 2 poles and one zero and the system is non-minimum phase system.

Question ID : **4410091565152**
Chosen Option : **2**

Q.77 A coil of 500 turns carries 2 A current. If the magnetic path length is 0.5 m and relative permeability of the core is 1000, what will be the value of magnetic field strength H?

Ans
1. 2000 A/m ✔
2. 1000 A/m ✘
3. 2000 Tesla ✘
4. 1000 Tesla ✘

Question ID : **4410091563526**
Chosen Option : **1**

Q.78 In the phasor diagram of a synchronous motor operating at lagging power factor, the excitation emf E leads the terminal voltage V by the load angle δ. If excitation is increased while load remains constant, what will the phasor diagram show?

Ans
1. δ increases and motor power factor worsens ✘
2. δ increases and motor power factor improves ✘
3. δ decreases and motor power factor improves ✔
4. δ decreases and motor power factor worsens ✘

Question ID : **4410091563574**
Chosen Option : **2**

Q.79 In electrostatic instruments, controlling torque is provided by:

Ans
1. Spring mechanism ✔
2. Permanent magnet ✘
3. Air vane ✘
4. Magnetic shunt ✘

Question ID : **4410091547683**
Chosen Option : **3**

Q.80 During the load flow analysis of an electrical power system, which quantities are unknown for a generator bus and must be found out by solving the power flow equations of the system?

(**Note:** P_i - net real power input at the bus,
Q_i - net reactive power input at the bus,
V_i - voltage magnitude at the bus,
δ_i - voltage phase angle at the bus)

Ans
1. $|V_i|$ and P_i ✘
2. δ_i and Q_i ✔
3. P_i and Q_i ✘
4. $|V_i|$ and δ_i ✘

Question ID : **4410091572701**
Chosen Option : **3**

Q.81 **A multiplexer can be used to convert ________**

Ans

1. Serial data to parallel data
2. Digital to analog data
3. Parallel data to serial data
4. Analog to digital data

Question ID : **4410091553661**
Chosen Option : **1**

Q.82 **If A, B, and C denote the minuend, subtrahend, and previous borrow respectively of a full subtractor, the Boolean expression for the borrow output is _______.**

Ans

1. A'B'C + A'BC' + AB'C' + ABC
2. A'B+BC+AC
3. A'B+BC+A'C
4. AB+BC+AC

Question ID : **4410091658350**
Chosen Option : **1**

Q.83 **Which logic gate would be used in a digital circuit to ensure power is supplied only when multiple switches are ON simultaneously?**

Ans

1. NOR gate
2. NOT gate
3. OR gate
4. AND gate

Question ID : **4410091792254**
Chosen Option : **4**

Q.84 **In fault analysis, the bus impedance matrix is significant because _______.**

Ans

1. It eliminates the requirement for symmetrical component analysis in fault current studies
2. It relates bus voltages to injected currents, enabling accurate fault current computation
3. It directly provides fault current values without requiring further numerical calculation
4. It remains unaffected by the electrical network configuration or overall system topology

Question ID : **4410091563848**
Chosen Option : **2**

Q.85 Given the $G(s)H(s) = \frac{k}{s(s+1)(s+3)}$, the point of intersection of the asymptotes of the root loci with the real axis is __________.

Ans
1. -3
2. 4
3. -1.33
4. -4

Question ID : **4410091661829**
Chosen Option : **1**

Q.86 The resolution of a 9-bit Ladder network digital to analog converter which produces a maximum analog output value of 5.11 volts is:

Ans
1. 5 mV
2. 10 mV
3. 20 mV
4. 5.11 V

Question ID : **4410091558288**
Chosen Option : **2**

Q.87 Two same frequency signals, with equal amplitudes and with some phase difference between them, are applied across the X and Y plates of a CRO. It was found that the pattern displayed on screen was an ellipse with its major axis lying in the 2^{nd} and 4^{th} quadrant. Which of the following can be a possible phase difference between these two signals?

Ans
1. 120°
2. 60°
3. 30°
4. 180°

Question ID : **4410091533313**
Chosen Option : **1**

Q.88 For a third-order system with the characteristic equation $s^3+as^2+bs+c=0$, what is the necessary and sufficient condition for stability?

Ans
1. a>0, b>0, c>0, and ab<c
2. a>0, b>0, c>0, and ac>b
3. a>0, b>0, c>0, and ab>c
4. a>0, b>0, c>0

Question ID : **4410091553459**
Chosen Option : **3**

Q.89 In the FM expression s(t) = Ac cos(2πfct + β sin(2πfmt)), what does the term β represent?

Ans
✔ 1. Modulation index
✘ 2. Carrier amplitude
✘ 3. Carrier frequency
✘ 4. Phase deviation

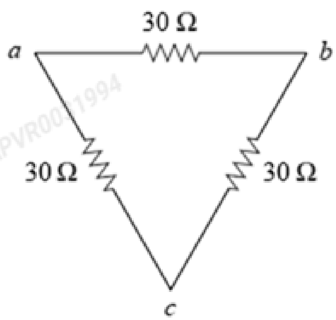
Question ID : **4410091807785**
Chosen Option : **1**

Q.90 An alternator's voltage regulation is often highest (positive and largest) while supplying ____________.

Ans
✘ 1. Unity power factor load
✔ 2. Lagging power factor load
✘ 3. Leading power factor load
✘ 4. No load

Question ID : **4410091545636**
Chosen Option : **1**

Q.91 For the figure given below showing delta connection, find the equivalent resistance per phase for the equivalent star connection.



Ans
✘ 1. 20 Ω
✘ 2. 90 Ω
✘ 3. 30 Ω
✔ 4. 10 Ω

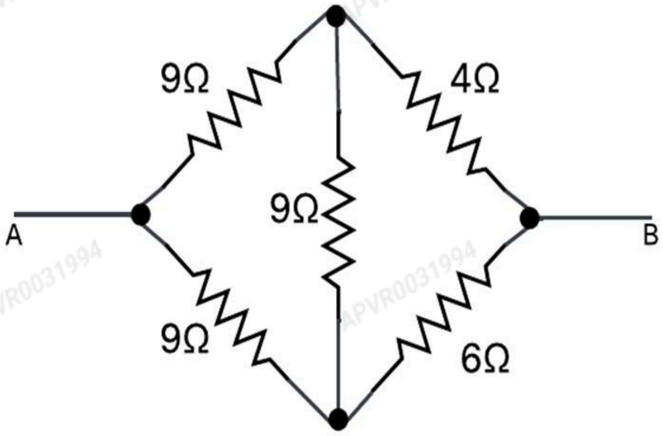
Question ID : **4410091554862**
Chosen Option : **2**

Q.92 Which of the following limits the high-frequency response of a BJT small-signal amplifier?

Ans
✘ 1. Coupling capacitors
✔ 2. Internal transistor capacitances
✘ 3. DC power supply
✘ 4. Bypass capacitors

Question ID : **4410091559091**
Chosen Option : **2**

Q.93 **Determine the equivalent resistance across terminals A and B in the given circuit.**



Ans

1. 4.187 Ω
2. 5.725 Ω
3. 6.937 Ω
4. 7.562 Ω

Question ID : **4410091565478**
Chosen Option : **1**

Q.94 **In a digital communication system, a shift register is used to convert 8 bits of serial data into parallel format. Which property of the D flip-flop makes it the most suitable choice for constructing this shift register?**

Ans

1. Its characteristic where the next state is exactly the present data input.
2. Its edge-triggered nature which prevents race conditions in the counter.
3. Its ability to toggle state when a control signal is high.
4. Its capability to hold data indefinitely as long as power is supplied.

Question ID : **4410091538982**
Chosen Option : **1**

Q.95 **Which of the following describes the role of the braking magnet when a single-phase energy meter is undergoing calibration?**

Ans

1. Adjust and control braking torque
2. Produce rotating magnetic field
3. Regulate supply voltage
4. It minimizes the effect of friction on the disc

Question ID : **4410091544749**
Chosen Option : **1**

Q.96 Which device is a voltage-controlled power transistor?

Ans
1. BJT
2. GTO
3. SCR
4. MOSFET

Question ID : **4410091563100**
Chosen Option : **1**

Q.97 In frequency modulation (FM), which parameter of the carrier signal remains constant during transmission?

Ans
1. Frequency of the carrier signal
2. Phase of the message signal
3. Amplitude of the carrier signal
4. Modulating frequency

Question ID : **4410091805686**
Chosen Option : **3**

Q.98 Per-unit impedance diagrams are used because they:

Ans
1. Vary with voltage level
2. Increase complexity
3. Are difficult to compute
4. Eliminate transformer turns ratio complications

Question ID : **4410091561223**
Chosen Option : **4**

Q.99 A quasi square wave inverter is fired at an angle of 30°. Calculate the ratio of 3rd harmonic component in the inverter output voltage with respect to Fundamental component in the output voltage.

Ans
1. 0
2. 0.11
3. 0.33
4. 0.20

Question ID : **4410091559876**
Chosen Option : **4**

Q.100 A quasi square wave inverter is fired in such a way that its 5^{th} harmonic component became zero in the output voltage. What is the value of firing angle (α) and thus the width (d) of the output voltage, respectively?

Ans

1. α = 36°, d = 108°

2. α = 36°, d = 144°

3. α = 30°, d = 120°

4. α = 18°, d = 144°

Question ID : **4410091559849**

Chosen Option : **2**

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